CLOSED SUBMODULES OF I-ADICALLY COMPLETE MODULES: (1) CLOSED IFF (2) N = INTERSECTION (N + I^N M) IFF (3) M/N COMPLETE; FINITE GENERATION GIVES N COMPLETE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file examples.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The author marks where finite generation is used ('this is where we use that I is finitely generated') - preserved verbatim.

Editorial note. The label lemma-closed-is-complete names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- I-adic completions N^ of submodules and density of N in N^; factorization of continuous maps into complete modules
- Formal Spaces, closed lemma for (1) iff (2); Algebra, quotient-complete lemma for (2) iff (3); Algebra, hathat-finitely-generated lemma for direct summands of complete modules

Source reference. Original lemma 05GG: sources/stacks-chow/tags.txt, source lines 7060–7060. Prerequisite pointer invoked at 836; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 031A: sources/stacks-chow/tags.txt, source lines 3977–3977. Prerequisite pointer invoked at 824; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AMS: sources/stacks-chow/tags.txt, source lines 13415–13415. Prerequisite pointer invoked at 822; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $I \subset R$ be an ideal of a ring. Let M be an I-adically complete R-module and let $N \subset M$ be a submodule. The following are equivalent*

(1) *N is closed in M,*
(2) *$N = \bigcap_{n \geq 1}(N + I^n M)$,*
(3) *M/N is I-adically complete*

If I is finitely generated, these conditions imply that N is I-adically complete.

Date: stacks-project:examples.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. The equivalence of (1) and (2) follows from Formal Spaces, Lemma 0AMS. The equivalence of (2) and (3) is Algebra, Lemma 031A. Assume I is finitely generated and (1) holds. Let $N^\wedge$ be the I-adic completion of N. Since M is I-adically complete we have a factorization

$$N \to N^\wedge \to M$$

of the inclusion map. Since $N^\wedge \to M$ is continuous and N is dense in $N^\wedge$ and N is closed in M, we see that $N^\wedge \to M$ factors through N. Thus $N^\wedge = N \oplus C$ for some A-module C. Thus N is a direct summand of the I-adically complete module $N^\wedge$, see Algebra, Lemma 05GG (this is where we use that I is finitely generated), and hence N is I-adically complete. $\square$